

From Production to Perception: Designing a Sociophonetic Study

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Research Questions

- RQ1** Can listeners identify the intended gay guise in a paired comparison?
RQ2 Does identification accuracy vary by listener group (L1 vs. L2)?
RQ3 Does the magnitude of the acoustic contrast predict identification accuracy?
RQ4 Do trait ratings shift after participants learn the study concerns sexual orientation?

Stimulus Properties

5 speakers × 2 guises (gay/straight). Same speaker, same passage, different persona.

Pair	Speaker	Dialect	SO	Passage	Dur. d	COG d	/s/ n	Excerpt	Design Role
1	L9	Venezuelan	LGBTQ+	Narrative	0.66	0.90	16/16	10–12 s	Strong, both cue types
2	L7	Venezuelan	LGBTQ+	Science	0.04	0.09	19/19	11–14 s	Null control (near-zero contrast)
3	H9	Basque	Non-LGBTQ+	Narrative	0.08	1.41	16/17	13–14 s	COG-only (spectral)
4	H8	Sevillian	Non-LGBTQ+	Science	0.55	0.95	20/20	10–12 s	Strong, both cue types
5	L2	Sevillian	LGBTQ+	Science	2.69	1.89	7/7	7–9 s	Maximum contrast

Design balance: Dialect: 2 Sevillian, 2 Venezuelan, 1 Basque • SO: 3 LGBTQ+, 2 non-LGBTQ+ • Passage: 2 narrative, 3 science • Sibilant systems: aspiration/deletion (Sevillian), variable coda (Venezuelan), apicoalveolar with full retention (Basque)

Counterbalancing: Half of participants hear gay guise as Speaker A; half hear straight as Speaker A. Pair presentation order randomized within participant.

Two-Pass Survey Architecture

Pass 1 — Unprimed (no mention of sexual orientation)

Per pair: Audio A → Visual Analog Scale (VAS) trait ratings (5 dimensions, see below) → Audio B → VAS trait ratings (5 dimensions)

After all 5 pairs: Tiered metalinguistic awareness block (see below)

▼ Transition: SO Disclosure + option to withdraw ▼

Pass 2 — Post-Disclosure

Per pair: Audio A → SO identification (forced-choice) + confidence VAS → Audio B → same

Then: Re-rate VAS trait dimensions (5 per speaker) for priming shift comparison

Bipolar Visual Analog Scale (VAS) Trait Dimensions (0–100, start at 50)

Left Anchor (–)	Dimension	Right Anchor (+)
Poco agradable	Pleasantness	Muy agradable
Poco seguro/a de sí	Confidence	Muy seguro/a de sí
Poco educado/a	Education	Muy educado/a
Poco masculino/a	Masculinity	Muy masculino/a
Nada familiar	Familiarity	Muy familiar

Tiered Metalinguistic Awareness Block

Funnel structure: each tier narrows focus while documenting where attention goes unprompted vs. directed.

Tier	Type	What It Asks	Logic / Data
1	Free description	Open text: describe differences between voices	All participants. Code for spontaneous /s/ mention.
2	Feature checklist	Select which features varied: vowels, consonants, speed, intonation, accent, rhythm, clarity, voice quality	All participants. Randomized order. "No differences" anchored at bottom as exclusive option.
3	Consonant elaboration	Open text: describe consonant differences in more detail	Display logic: only if "consonantes" selected in Tier 2. Non-selection = data.
4	Dialect placement	Where is each speaker from? Dropdown per speaker with 8 region options	All participants. Indirect /s/ sensitivity measure (dialect ID relies on sibilant patterns).

Outcome Measures & Analysis

RQ	Dependent Variable	Key Predictors	Data Source
1	SO identification accuracy (Pass 2 forced-choice, recoded)	Acoustic contrast (Cohen's d), dialect	Pass 2
2	SO identification accuracy × listener group	L1 vs. L2, ATG score, LGBTQ+ familiarity	Pass 2
3	Identification accuracy × contrast magnitude	Pre-measured duration d + COG d per pair	Pass 2 + acoustic data
4	VAS trait shift (Pass 2 – Pass 1 per dimension)	Disclosure (within-subject), listener group	Pass 1 + Pass 2 VAS

Planned analysis: Cumulative link mixed-effects models (ordinal regression) with listener group, dialect, and acoustic contrast as fixed effects; random intercepts for participant and speaker.

Listener-Level Covariates

All Participants • L1 vs. L2 status • ATG scale (attitudes toward gay men) • LGBTQ+ familiarity index • Dialect familiarity ratings	L2 Participants Only • Self-rated proficiency level • Immersion experience (duration, location) • Formal instruction years • Dialect exposure history
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